

Proceedings Track

Truth Has a Boundary. A Machine-Verified Topological Constraint on Models That Separate Truth from Falsehood

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Abstract

Probing studies read truth off a language model's activations, treating truth as a continuous decodable field over representation space. We ask what the shape of that space requires of any model answering over it. Our central result is a coupling theorem. If a continuous model on a connected representation space ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query must carry confidence exactly at the decision threshold while its answer sits exactly on the truth boundary. The theorem mentions no safety guarantee, so it does not depend on how a guarantee treats its own threshold. A guarantee including its threshold is impossible, and one excluding it survives but is silent at the coupled query. Under Lipschitz regularity the coupled point sits inside an ambiguity tube of computable width, any feasible system must carry strictly positive truth-side slack, and for nonzero linear probes the truth boundary is an affine hyperplane of codimension one. A symmetry version needs no coverage assumption, a discrete analysis prices the removal of topology, and multi-turn and stochastic extensions follow, with falsehood derived in the stochastic dichotomy. Every theorem is machine-verified in Lean 4 against Mathlib with no sorry placeholders, the main results reducing to the three standard kernel axioms. An appendix observes the predicted coupling in five small language models, with the model's own confidence signal and against null baselines, and an adversarial attempt to train away the slack plateaus strictly above zero.

Keywords: representational geometry, topology, uncertainty, calibration, intermediate value theorem, Borsuk-Ulam, formal verification, Lean

1. Introduction

Truth is geometrically simple inside language models. Linear probes separate true from false statements in activation space, latent knowledge can be found without supervision, and steering along learned directions makes models more truthful (Marks and Tegmark, 2024; Azaria and Mitchell, 2023; Burns et al., 2023; Li et al., 2023; Zou et al., 2023). All of this work treats truth as a continuous, decodable field over representation space.

We take that picture seriously as geometry and ask what follows. Suppose truth is a continuous scalar field on a connected representation space, and the model's answers and confidences are continuous fields on the same space. These assumptions mention no architecture, training, or scale, applying to any model whose fields are defined and continuous on the stated domain. What they yield is a coupling theorem. If the model ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query carries confidence exactly at the threshold while its answer sits exactly on the truth boundary.

The proof is short, and the theorem's value lies in what it does not mention, no safety guarantee and no threshold convention, so every policy question becomes a corollary. An inclusive guarantee is impossible, an exclusive one survives but says nothing at the point

Proceedings Track

topology guarantees, and slack buys safety. Under Lipschitz regularity the coupled point sits inside an ambiguity tube of computable width, and for the nonzero linear probes used in practice the truth boundary is an affine hyperplane.

Contributions. The boundary coupling theorem with its corollary taxonomy (Section 3); the geometry of the boundary, closed, crossed by every true-to-false path, a hyperplane for nonzero linear probes, with an ambiguity tube of strictly positive truth-side slack (Section 4); a symmetry version needing no coverage (Section 5); a discrete analysis with multi-turn and stochastic extensions (Section 6); and full machine verification (Section 6).

The mathematics is elementary, and the contribution is not technical difficulty. It is a framing in which exact threshold uncertainty is a topological property of representation space, precision, since the theorems name every assumption in play, and verification, since every statement is machine-checked, leaving the premises as the only thing to argue about. The theorems do not say models err often and assign no probability to failure. The coupled point is neither strictly true nor strictly false, and derived falsehood appears only in the stochastic dichotomy.

Related work. Statistical and computability arguments make hallucination inevitable in other senses (Kalai and Vempala, 2024; Kalai et al., 2025; Xu et al., 2024); ours is topological and yields uncertainty rather than confident falsehood. The probing literature models truth as a continuous decodable field (Marks and Tegmark, 2024; Burns et al., 2023; Azaria and Mitchell, 2023; Li et al., 2023; Zou et al., 2023; Park et al., 2024), exactly our hypothesis on δ , and our results give that program a frontier. Connected-domain modeling follows the manifold hypothesis (Fefferman et al., 2016; Cai et al., 2021), and the symmetry section is a scalar relative of Borsuk-Ulam (Borsuk, 1933; Matoušek, 2003). To our knowledge this is the first fully machine-verified impossibility-style result on model truthfulness at the level of representation geometry (de Moura and Ullrich, 2021; mathlib Community, 2020). Appendix A extends all four threads.

2. The setting

$\mathcal{X}$ is a topological space of queries as the model represents them, prompt embeddings or residual-stream states, $\mathcal{A}$ a topological space of answers, and our standing assumption is simple.

(T) $\mathcal{X}$ is connected and nonempty.

Ambient embedding spaces are standardly modeled as $\mathbb{R}^d$ or connected submanifolds, which satisfy (T), though the reachable set of a token-driven model may be disconnected. When guarantees range over the ambient space, as they do whenever a probe or steering vector applies at arbitrary points, (T) holds. Continuity likewise idealizes trained networks, softmax and embeddings being continuous while argmax decoding is not, and Section 6 prices dropping either idealization.

Definition 1 (Model and truth fields) A model is a map $F = (a, c)$ from $\mathcal{X}$ to $\mathcal{A} \times \mathbb{R}$, with answer $a(x)$ and scalar confidence $c(x)$, continuous if both components are. A truth field is a continuous δ from $\mathcal{X} \times \mathcal{A}$ to $\mathbb{R}$, a signed truth-distance, negative on strictly true

Proceedings Track

answers, positive on strictly false ones, with the truth boundary as its zero set. The realized truth field is $\delta_F(x) = \delta(x, a(x))$.

A linear probe $w^\top x - b$ on activations (Marks and Tegmark, 2024; Burns et al., 2023) is a candidate continuous δ_F , its zero set the probe’s classification boundary. Whether that coincides with semantic truth away from the fitted samples is an empirical premise, not a theorem, and our results hold under it.

Definition 2 (Truth-separating at threshold τ) Fix a threshold τ , a truth slack $\varepsilon \geq 0$, and a confidence margin $\gamma \geq 0$. F is (ε, γ) -separating at τ if $\delta_F(x) < -\varepsilon$ implies $c(x) > \tau + \gamma$ and $\delta_F(x) > \varepsilon$ implies $c(x) < \tau - \gamma$. The slack is in units of δ_F and the margin in units of c , so the condition survives independent rescaling. Exactly separating means both are zero.

Definition 3 (Covering) F is ε -covering if some query has $\delta_F < -\varepsilon$ and some has $\delta_F > \varepsilon$, at $\varepsilon = 0$ somewhere strictly right and somewhere strictly wrong.

Definition 4 (Guarantees) A threshold-inclusive guarantee says that $c(x) \geq \tau$ implies $\delta_F(x) < 0$, and a threshold-exclusive one that $c(x) > \tau$ implies $\delta_F(x) < 0$.

Covering is satisfied by any model ever right and ever wrong, separation idealizes a perfect truth probe used as confidence, the guarantees idealize mitigation, the coupling results hold at any threshold, and the biconditional variants sit at one half.

Remark 5 (This is not statistical calibration) Separation is pointwise and much stronger than statistical calibration (Guo et al., 2017), and one-way by construction. The biconditional variant adds the converses, so the sign of δ_F and the side of the threshold determine each other. The symmetry and discrete results use it, and it implies the one-way pair, Lean `strictCalibrated_to_epsCalibrated_zero_half`.

3. The boundary coupling theorem

Topology enters through one theorem needing no confidence.

Theorem 6 (Boundary existence) Let $\mathcal{X}$ satisfy (T), the answer map and truth field be continuous, and F be 0-covering. Then some x_0 has $\delta_F(x_0) = 0$, since δ_F is continuous with both signs and the IVT applies on the connected space. Lean `model_truth_boundary_nonempty`, IVT step `IsPreconnected.intermediate_value2`.

Theorem 7 (Boundary coupling) Let $\mathcal{X}$ satisfy (T). Let c be continuous, and let F be 0-covering and exactly separating at τ . Then some x_0 satisfies both

$$c(x_0) = \tau \quad \text{and} \quad \delta_F(x_0) = 0.$$

Proof Separation converts the coverage witnesses into confidence witnesses on both sides of τ , and the IVT gives x_0 with $c(x_0) = \tau$. Were $\delta_F(x_0)$ negative, separation would give

Proceedings Track

$c(x_0) > \tau$, and positive, $c(x_0) < \tau$, so $\delta_F(x_0) = 0$. This is Lean theorem `boundary_coupling`. ■

The theorem mentions no guarantee, so no threshold convention affects it. We call such an x_0 a *coupled point*, and what it means for safety is policy, with each policy a corollary.

Corollary 8 (Policy taxonomy) *Adding a threshold-inclusive guarantee to Theorem 7’s hypotheses is jointly contradictory, the trilemma, since coverage, exact separation, and an inclusive guarantee cannot jointly hold. Adding a threshold-exclusive guarantee yields no contradiction, but the coupled point satisfies $c(x_0) = \tau$, its antecedent fails exactly there, and the system owns a boundary query about which its guarantee says nothing. Lean theorems `closed_guarantee_impossible`, `hallucination_trilemma`, and `open_guarantee_silent`.*

Defining the guarantee with a strict inequality therefore does not dissolve the impossibility. The coupling theorem is convention-free, the convention only selects which clause of the taxonomy applies, and the coupled point exists under either convention.

4. The geometry of the boundary

Theorem 9 (Closed, and crossed by every path) *For any topological $\mathcal{X}$ and $\mathcal{A}$, continuous δ and answer map, $\{x \mid \delta_F(x) = 0\}$ is closed in $\mathcal{X}$, and any continuous path from a query answered strictly truly to one answered strictly falsely contains a point exactly on the truth boundary. Lean theorems `model_truth_boundary_isClosed` and `path_crosses_truth_boundary`.*

Theorem 10 (Linear probes have hyperplane boundaries) *Let f be a nonzero linear probe on a d -dimensional real space with offset b . Then $\{x \mid f(x) = b\}$ is an affine translate of the kernel of f , of dimension exactly $d - 1$. Lean theorems `linear_probe_boundary_coset` and `linear_probe_boundary_dim`, by rank-nullity.*

The coupling theorem therefore places a threshold-confidence answer on a codimension-one affine hyperplane. Next we quantify what slack changes.

Theorem 11 (Corridor and tube) *Let c be L_c -Lipschitz and δ_F be L_δ -Lipschitz. For $\gamma > 0$, queries with $c \geq \tau + \gamma$ and with $c \leq \tau - \gamma$ are at distance at least $2\gamma/L_c$, and for a coupled point x_0 , every query within distance r of x_0 has truth margin at most $L_\delta r$ and confidence within $L_c r$ of the threshold. Lean theorems `confidence_gap_width`, `ambiguity_tube`, and `truth_margin_tube`.*

The coupled point is therefore the center of a tube of near-ambiguous queries, and decisive regions cannot approach closer than the corridor width.

Theorem 12 (Approximate coupling, and slack must be positive) *Let $\mathcal{X}$ satisfy (T), let c be continuous, and let F be ε -covering, (ε, γ) -separating at τ , and carry a threshold-inclusive guarantee. Then some x_0 has $c(x_0) = \tau$ and $-\varepsilon \leq \delta_F(x_0) < 0$, and consequently the truth slack satisfies $\varepsilon > 0$ for every margin $\gamma \geq 0$, so zero truth slack is infeasible. These are Lean theorems `two_slack_approx_coupling`, `exact_from_approx`, and `truth_slack_must_be_positive`.*

Proceedings Track

Slack is therefore a geometric budget, positivity attaching to the truth side alone. We read the infimum of feasible truth slack as the system’s truth-boundary width, with the relative width of Appendix D as a dimensionless proxy, and under the guarantee no query with confidence between τ and $\tau + \gamma$ sits exactly on the boundary (Lean `no_boundary_in_upper_band_two_slack`).

5. Symmetry replaces coverage

Symmetry can replace coverage. Suppose semantic negation acts on $\mathcal{X}$ as a continuous involution σ with no fixed point and the realized truth field is odd under it, idealizing the measured reflection of truth directions by negation (Marks and Tegmark, 2024).

Theorem 13 (Odd fields vanish) *Let $\mathcal{X}$ satisfy (T) with such an action σ , and let g be continuous and odd under σ . Then g has a zero, by the IVT between any x and σx . This is Lean theorem `antipodal_odd_has_zero`.*

This is the scalar shadow of Borsuk-Ulam, needing no sphere. Two refinements make it robust and concrete.

Theorem 14 (Robust and concrete oddness) *Let $\mathcal{X}$ satisfy (T), σ map $\mathcal{X}$ to itself, and g be continuous with $|g(\sigma x) + g(x)| \leq \varepsilon$ for all x . Then some x has $|g(x)| \leq \varepsilon/2$. Moreover, on the unit sphere of $\mathbb{R}^d$, $d \geq 2$, with negation the antipodal map, every continuous odd field vanishes somewhere on the sphere. Lean theorems `approx_odd_near_zero` and `sphere_odd_has_zero`.*

The near-boundary point degrades linearly with the defect, and the sphere is, up to a learned affine map, where layer normalization places its states, negation the candidate antipode.

Theorem 15 (Antipodal coupling) *Let $\mathcal{X}$ satisfy (T) with a free continuous involution σ , F be continuous with biconditional separation at $\frac{1}{2}$, and δ_F be odd. Then some x has $c(x) = \frac{1}{2}$ and $\delta_F(x) = 0$. Lean `antipodal_hallucination_trilemma`.*

No coverage hypothesis appears, so a negation-closed family of questions meets the truth boundary by symmetry alone. Equivariance, usually a design virtue (Bronstein et al., 2021), is here the engine of the obstruction.

6. Leaving the continuum, adding turns and randomness

Token spaces are finite, and one may object that connectedness does illegitimate work. The discrete analysis prices the objection, since without topology the boundary witness must be assumed rather than derived.

Theorem 16 (Discrete impossibility) *Let $\mathcal{X}$ and $\mathcal{A}$ be arbitrary sets with arbitrary F and δ , with biconditional separation at $\frac{1}{2}$. If some query has $\delta_F = 0$, a threshold-inclusive guarantee fails, and such a guarantee is possible exactly when no boundary query exists. These are Lean theorems `boundary_question_impossible` and `faithful_iff_boundary_free`.*

Proceedings Track

The assumed witness is what connectedness derived free in Theorem 6, and without it a crossing survives.

Proposition 17 (Discrete sign change) *Let g map $\{0, \dots, n\}$ to $\mathbb{R}$ with $g(0) < 0 < g(n)$. Then some i has $g(i) < 0 \leq g(i+1)$. This is Lean theorem `discrete_sign_change`.*

Along any token path from true to false, δ_F crosses zero between adjacent steps, a straddling pair rather than a boundary state, hence no contradiction alone. The continuum upgrades the crossing into a witness. Interaction does not escape, since $\mathcal{X}^T$ is connected and the arguments transport verbatim (`multi_turn_history_dependent`). Nor does randomization. On the expected fields the coupling theorem yields expected confidence one half at expected truth-distance zero (`stochastic_coupling_expected`), and a dichotomy converts expectation into sampling behavior.

Theorem 18 (Stochastic dichotomy) *Let (Ω, μ) be a probability space and let c, d be real functions on Ω with d integrable and of mean zero. Suppose sample faithfulness holds almost everywhere, meaning $c \geq \frac{1}{2}$ implies $d < 0$, and suppose $\mu\{c \geq \frac{1}{2}\} > 0$. Then $\mu\{d > 0\} > 0$. This is Lean theorem `stochastic_dichotomy`.*

A model faithful on almost every sample and confident with positive probability at a coupled point must emit strictly false answers with positive probability. Randomization spreads the boundary across samples, and here falsehood is derived.

Machine verification. Every theorem and proposition corresponds to a named, sorry-free Lean 4 result (de Moura and Ullrich, 2021; mathlib Community, 2020). The artifact builds with zero errors, axiom audits reduce each main theorem to exactly the three standard kernel axioms, and Table 1 maps every result to its identifier, `hallucination_master_theorem` bundling the core impossibility. The prose claims nothing beyond what is formalized.

7. Discussion

The theorem rests on connectedness, continuity, coverage, and separation, the trilemma adding an inclusive guarantee. Each hypothesis can be relaxed, each relaxation is a design stance, and each carries a named price, developed in Appendix A. Refusal training is connectedness surgery, restricting to reachable queries is the same move while probing and steering deploy over the ambient space, giving up coverage is vacuous for general systems, and dropping continuity keeps only the adjacent crossing. The recommended stance is Theorem 12's, separate to a band, with the infimum of feasible truth slack as a measurable diagnostic.

Limitations and outlook. The results are existence statements, so nothing follows about error rates, boundary points are the deterministic conclusion with Theorem 18 the stochastic exception, and the truth field idealizes probe behavior, Theorem 14 pricing oddness. Appendix D reports empirical checks on five small models, and next steps are higher Borsuk-Ulam machinery and nonlinear probe boundaries (Engels et al., 2025), in brains and machines alike. The geometry of representation space can prove theorems, and machines can check them.

Proceedings Track

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Proceedings Track

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Appendix A. Extended discussion and related work

Hallucination inevitability. Kalai and Vempala (2024) show statistically that calibrated models must hallucinate on facts seen once in training, Kalai et al. (2025) extend this line, and Xu et al. (2024) give a computability-theoretic inevitability argument. Our mechanism is topological, lives in representation space, localizes the conclusion at the truth boundary, and is more modest, since the deterministic theorems yield uncertainty rather than confident falsehood.

Proceedings Track

Geometry of truth in representations. Linearly decodable truth (Marks and Tegmark, 2024), unsupervised latent-knowledge discovery (Burns et al., 2023), lie detection from internal states (Azaria and Mitchell, 2023), and causal steering along truth directions (Li et al., 2023; Zou et al., 2023) all model truth as a continuous decodable field on activation space, and the linear representation hypothesis formalizes concepts as directions in that space (Park et al., 2024). Confidence signals (Farquhar et al., 2024; Kadavath et al., 2022) are treated similarly. Topological analyses of networks (Naitzat et al., 2020; Hensel et al., 2021) and geometric deep learning (Bronstein et al., 2021) engineer or analyze such structure, and we derive constraints from it.

Disconnect the space. Abstention breaks the argument at the IVT step, since declining to answer on an open region deletes it from the domain and can disconnect the true region from the false region. Refusal training, on this reading, is connectedness surgery. Under the symmetry version the cut must separate every representation from its negation image, so piecemeal refusal is not enough.

Quantify only over reachable queries. The coupled point may be an ambient activation that no real prompt produces, and a guarantee quantified only over reachable representations is then untouched. The objection is fair, and it is the domain version of disconnecting the space. But probing and steering pipelines apply their readouts at interpolated and edited activations, which is deployment over the ambient space, and a guarantee over the uncertified reachable set inherits every unknown of that set. Restricting the domain is a coherent stance if stated as one.

Give up coverage or continuity. A model that never answers strictly falsely escapes. That is vacuous for general-purpose systems and meaningful for retrieval-grounded ones restricted to verified content. Dropping continuity also escapes, at the price computed in Section 6, where only the adjacent-crossing statement survives unconditionally.

Implications for probing and steering. Truth-probe pipelines posit a continuous δ_F , and steering moves representations relative to its zero set (Li et al., 2023; Zou et al., 2023). Completing such a pipeline over a connected domain with both truth values realized yields an exact-uncertainty point on the probe’s own boundary, a hyperplane for linear probes, and mitigation effort should go there.

Appendix B. Formal statement conventions

The Lean formalization states results over arbitrary topological spaces Q and A , with a `ConnectedSpace` instance on Q , a model M from Q to $A \times \mathbb{R}$, and a truth field `delta` on $Q \times A$ (rendered here in ASCII). The realized truth field is

```
fun q => delta (q, (M q).1)
```

and confidence is `(M q).2`. The main conditions are named as follows, and some identifiers retain historical names.

- `TwoSlackSeparating M delta c eps gamma` is separation at threshold c with truth slack eps and confidence margin gamma . The equal-slack case is `EpsCalibrated M delta c eps`, definitionally `(epsCalibrated_iff_twoSlack)`.

Proceedings Track

- Covering with slack ϵ is `EpsCovering M delta  $\epsilon$` .
- The biconditional variant at threshold one half is `StrictCalibrated M delta`.
- The threshold-inclusive guarantee at threshold one half is `TrilemmaFaithful M delta`.
The general threshold version appears inline in the theorems that use it.

The embedding-space instantiation uses `EuclideanSpace R (Fin d)`. The quantitative results use `PseudoMetricSpace` and explicit Lipschitz hypotheses, and the hyperplane results use `LinearMap` with rank-nullity.

Table 1: Every result in the paper and its machine-checked counterpart.

Result	Lean identifier
Thm. 6	<code>model_truth_boundary_nonempty</code>
Thm. 7	<code>boundary_coupling</code>
Cor. 8	<code>closed_guarantee_impossible, hallucination_trilemma, open_guarantee_silent</code>
Thm. 9	<code>model_truth_boundary_isClosed, path_crosses_truth_boundary</code>
Thm. 10	<code>linear_probe_boundary_coset, linear_probe_boundary_dim</code>
Thm. 11	<code>confidence_gap_width, ambiguity_tube, truth_margin_tube</code>
Thm. 12	<code>two_slack_approx_coupling, exact_from_approx, truth_slack_must_be_positive</code>
Upper-band exclusion	<code>no_boundary_in_upper_band_two_slack</code>
Thm. 13	<code>antipodal_odd_has_zero</code>
Thm. 14	<code>approx_odd_near_zero, sphere_odd_has_zero</code>
Thm. 15	<code>antipodal_hallucination_trilemma</code>
Thm. 16	<code>boundary_question_impossible, faithful_iff_boundary_free</code>
Prop. 17	<code>discrete_sign_change</code>
Multi-turn	<code>multi_turn_history_dependent</code>
Stochastic expected-value coupling	<code>stochastic_coupling_expected</code>
Thm. 18	<code>stochastic_dichotomy</code>
Bundled master theorem	<code>hallucination_master_theorem</code>

Appendix C. Core proof term

The Lean proof of the coupling theorem, lightly annotated and rendered in ASCII.

```
theorem boundary_coupling
  {Q A : Type*} [TopologicalSpace Q] [ConnectedSpace Q]
  [TopologicalSpace A]
  (M : Q -> A x R) (delta : Q x A -> R) (c : R)
  (hconf : Continuous (fun q => (M q).2))
  (hcov : EpsCovering M delta 0)
  (hcal : EpsCalibrated M delta c 0) :
  exists q0, (M q0).2 = c /\ delta (q0, (M q0).1) = 0 := by
```

Proceedings Track

```

obtain <<qt, ht>, <qf, hf>> := hcov
-- separation turns coverage witnesses into confidence witnesses
have hct : (M qt).2 > c + 0 := hcal.1 qt ht
have hcf : (M qf).2 < c - 0 := hcal.2 qf hf
have hct' : (M qt).2 > c := by linarith
have hcf' : (M qf).2 < c := by linarith
-- IVT on the confidence field
obtain <q0, _, hq0> :=
  isPreconnected_univ.intermediate_value2
    (mem_univ qf) (mem_univ qt)
    hconf.continuousOn continuous_const.continuousOn
    (le_of_lt hcf') (le_of_lt hct')
refine <q0, hq0, ?_>
-- at conf = c, separation excludes both signs
by_contra hne
rcases lt_or_gt_of_ne hne with hlt | hgt
. have hd : delta (q0, (M q0).1) < -0 := by simpa using hlt
  have := hcal.1 q0 hd
  linarith
. have := hcal.2 q0 hgt
  linarith

```

Appendix D. Empirical illustration on small language models

The theorems are about idealized fields, so we checked whether their predictions are visible in real models. We used five open language models, namely GPT-2 with 124M parameters, Pythia-410M, Qwen2.5-0.5B, Phi-1.5 with 1.3B parameters, and Qwen2.5-1.5B, the first three below one billion parameters. The data are the matched city statements and their negations from the geometry-of-truth datasets of [Marks and Tegmark \(2024\)](#), giving 1496 true and false statement pairs such as “The city of Paris is in France” and its negation.

Setup. We embed each raw statement at its final token and select the layer whose probe has the best validation accuracy. The truth field δ_F is a linear logistic probe trained on one split of cities. The confidence field c is a second linear logistic head trained on a disjoint split with a different seed, so the two fields are independently trained readouts of the same representation space. This is the canonical construction the theory models, since a probing pipeline that uses a truth readout as a confidence signal is exactly a truth-separating c . We then interpolate hidden states between each matched pair, from the true statement to its negation, and record both fields along each path. We keep the paths whose endpoints the probe classifies correctly, which is 83 paths for GPT-2, 104 for Pythia-410M, 137 for Qwen2.5-0.5B, 66 for Phi-1.5, and 148 for Qwen2.5-1.5B. Test cities are held out from both training splits.

Findings. Table 2 and Figure 1 summarize. First, the premises hold approximately. Linear probes separate truth with accuracy 0.72 to 1.00, and only the two Qwen models are well above chance when asked the question verbally, at 0.71 and 0.96 against 0.48 to

Proceedings Track

0.56 for the rest, which replicates the known gap between represented and verbalized truth. Second, every kept path shows the adjacent sign change predicted by the discrete result. Third, confidence is pinned near the threshold at the boundary. The mean value of c at the probe’s zero crossing is 0.52, 0.51, 0.49, 0.43, and 0.54 for the five models, against decisive values near 0 and 1 at the endpoints. Fourth, the two independently trained fields couple. The median gap between the zero of δ_F and the threshold crossing of c tracks probe accuracy monotonically across all five models, from 0.31 of the path for Phi-1.5, whose probe is weakest, to 0.05 for Qwen2.5-1.5B, whose probe is strongest. The ordering follows separation quality and not parameter count alone, since Phi-1.5 outweighs every model except Qwen2.5-1.5B yet is the weakest separator on this domain. A sweep of all 24 of its layers finds no linear probe above 0.73, so the weakness is a property of the model on this domain and not of the layer choice. The two Qwen models give a within-family scale comparison, and there scale helps through separation quality, raising the probe from 0.96 to 1.00 and halving the gap from 0.10 to 0.05. Fifth, the relative boundary width, the median $|\delta_F|$ over queries whose confidence lies in $[0.4, 0.6]$ divided by the endpoint scale, runs from 0.37 for the weakest probe to 0.11 for the strongest. Sixth, the kNN graph of held-out activations places true and false statements in one shared main component for all five models, so truth values are not separated into different components, though three models show a second component. A diagnostic below identifies what that component is.

The model’s own confidence. The design above trains both fields on truth, so nearby boundaries could reflect shared fitting. We therefore repeat the experiment with a confidence field the model supplies itself, with no trained head, read out through unembedding weights fixed by pretraining before our data existed. We capture the final block’s last-token state under a question template, interpolate between matched pairs as before, patch the interpolated state back into the final block, and read the model’s own probability of answering true over the two answer tokens. The truth probe is trained as before on a disjoint split of the same states. The question-template states change which endpoint pairs the probe classifies correctly, giving 53, 84, 137, 80, and 147 kept paths here. The endogenous field satisfies the separation premise on a path when it is decisive at both endpoints, meaning its confidence lies on the correct side of one half at both. Table 3 and Figure 2 report what follows. In Qwen2.5-1.5B the premise holds on 146 of 147 paths with median gap 0.08, and in Qwen2.5-0.5B on 128 of 137 paths with median gap 0.13, both against null gaps of 0.40 and above. In Pythia-410M the premise fails on every path, since the model answers true regardless of the statement, and the theory predicts nothing. GPT-2 sits between, with 23 of 53 paths decisive and a gap that does not separate from the random-head null. Phi-1.5 is endpoint decisive on 53 of 80 paths, but its verbal accuracy is 0.56 and its gap stays at null level, so endpoint decisiveness without verbal capability does not amount to separation. Coupling with the model’s own uncertainty appears cleanly where the separation premise holds throughout, and never where it fails.

Null baselines. To calibrate the gaps, we replace the trained confidence head with heads trained on permuted labels and with random directions, twenty seeds each, evaluated on the same kept paths as the trained heads (Table 3). The trained-head gaps of 0.25, 0.20, 0.10, 0.31, and 0.05 fall below the null medians for all five models, and below the entire null range for Pythia-410M and both Qwen models. For GPT-2 and Phi-1.5 the observed gaps

Proceedings Track

lie inside the random-head null range, so the trained-head evidence is weak there on its own. Bootstrap intervals over paths, with ten thousand replicates, make the gaps precise. The 95 percent intervals for the median trained-head gap are 0.175 to 0.325, 0.15 to 0.287, 0.10 to 0.125, 0.175 to 0.375, and 0.05 to 0.075 across the five models, with 0.10 to 0.15 for Qwen2.5-0.5B’s endogenous field and 0.075 to 0.10 for Qwen2.5-1.5B’s. Every interval upper bound sits below the corresponding null median except the endogenous cases of GPT-2 and Phi-1.5.

A second dataset. The same pipeline runs on the Spanish-English translation statements from the same source, 354 matched pairs, with splits by statement since no city structure exists. Both Qwen models replicate the cities result, with probe accuracies 0.98 and 0.94 and median crossing gaps 0.05 and 0.10. Pythia-410M’s probe drops to 0.70 on this harder set and its gap rises to null level, and Phi-1.5’s probe reaches 0.71 with its gap at null level too. GPT-2’s probe is at chance there, with too few valid paths to score. Coupling again appears exactly where separation holds.

What the second component is. The kNN check deserves one more look, since three models showed a second component holding half the points. The diagnostic answer is clean. The components split exactly by surface form. One holds the positive statements, the other the negations, and each contains true and false statements in equal measure. What separates activations is syntax, not truth, and both truth values coexist inside each component. The interpolation paths are line segments in the ambient space, which are connected regardless of how the data clusters.

Measuring the oddness defect. The symmetry section rests on negation acting oddly on the truth field, and the matched pairs measure how odd it is. The defect $|\delta_F(x) + \delta_F(\sigma x)|$ relative to the endpoint scale has median 0.40, 0.43, 0.30, 0.53, and 0.17 for GPT-2, Pythia-410M, Qwen2.5-0.5B, Phi-1.5, and Qwen2.5-1.5B, smallest for the strongest model. Exact oddness fails in real representations, which is why Theorem 14 carries the approximate version. Its near-boundary point survives any defect, and empirically every kept path crosses zero regardless. The width column of Table 2 also gives the slack diagnostic of Section 4 an empirical proxy, between 0.11 and 0.37 of the endpoint scale on these models.

Nonlinear probes. The linear truth field crosses zero along a linear path by construction, so we replace both fields with one-hidden-layer MLPs trained on the same splits. Nothing about the path behavior is then guaranteed, the boundary is no longer a hyperplane, and multiple crossings become possible. The nonlinear probes reach 0.79, 0.87, 0.97, 0.80, and 1.00, and give 124, 121, 144, 104, and 149 kept paths. Every kept path still crosses zero, and at least 98 percent of paths in every model cross exactly once, so the nonlinear boundary is still met transversally along these paths. The coupling tightens or holds in every model, with median gaps 0.15, 0.20, 0.075, 0.175, and 0.05, and confidence at the crossing averaging 0.42, 0.50, 0.42, 0.38, and 0.49. The coupling is therefore not an artifact of linearity.

An adversarial falsification attempt. The strongest test of an impossibility result is a maximally empowered attempt to violate it. We freeze the truth field and train a confidence head directly on the forbidden objective, confidence above one half wherever the field says true and below one half wherever it says false or boundary, with the guarantee side weighted

Proceedings Track

five times harder. The adversary trains on all non-test statements plus fresh continuum interpolants every epoch, and is evaluated on 21,000 to 30,000 held-out path points per model. Theorem 12 predicts three things. The achieved truth slack must stay positive, the residual failures must localize at the boundary, and slack must trade against guarantee violations. All three appear. The achieved slack falls and then plateaus between 0.22 and 0.91 of the endpoint scale across the five models, exactly flat over the final fifteen epochs for Qwen2.5-0.5B, and never reaches zero. The median depth of failing points shrinks from 0.40 or more to between 0.04 and 0.11, so failures concentrate in the tube around the boundary. And the guarantee stays violated on a small fraction of points throughout. The theorem forbids only exactly zero slack for any fixed continuous head, so a stronger adversary could push the floor lower, and Theorem 11 prices that escape, since a sharper transition requires a proportionally larger Lipschitz constant. The observed plateau reflects the floor set by this adversary’s regularity, and the localization shows what sharpening buys, a thinner tube around a boundary that does not go away.

What this does and does not show. The truth field of the main design is linear, so its sign change along a linear path is guaranteed analytically, and the nonlinear replication above removes exactly that guarantee. The empirical content lies elsewhere. The premises hold to varying degrees in real small models. The two fields are trained on disjoint data with different seeds, so their coupling at the boundary is not an artifact of shared training data, though both target truth. The endogenous experiment addresses the residual concern where the model itself can verbalize truth. The pinned confidence value lands at one half. And the boundary width is a measurable quantity that differentiates models. Code, results, and the figure are included in the supplementary material.

Table 2: Boundary coupling in five small models. Layer is the selected layer. Probe and Head are held-out accuracies of the truth probe and the confidence head. Verbal is the model’s own true or false answer accuracy. Cross is the mean confidence at the probe’s zero crossing. Gap is the median distance between the two fields’ crossings as a fraction of the path. Width is the relative boundary width.

Model	Layer	Probe	Head	Verbal	Cross	Gap	Width
GPT-2 (124M)	9/12	0.77	0.72	0.48	0.52	0.25	0.33
Pythia-410M	18/24	0.82	0.77	0.50	0.51	0.20	0.36
Qwen2.5-0.5B	12/24	0.96	0.96	0.71	0.49	0.10	0.20
Phi-1.5 (1.3B)	18/24	0.72	0.72	0.56	0.43	0.31	0.37
Qwen2.5-1.5B	14/28	1.00	1.00	0.96	0.54	0.05	0.11

Appendix E. Artifact checklist

The anonymized supplementary zip contains the complete Lean project and the experiment. The Lean part includes the build configuration, the toolchain pin for Lean v4.28.0, the manifest pin for Mathlib v4.28.0, and seventeen source files. To reproduce, run `lake build`. It completes with zero errors and no `sorry`. Axiom sets of the main theorems are printed by

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Proceedings Track

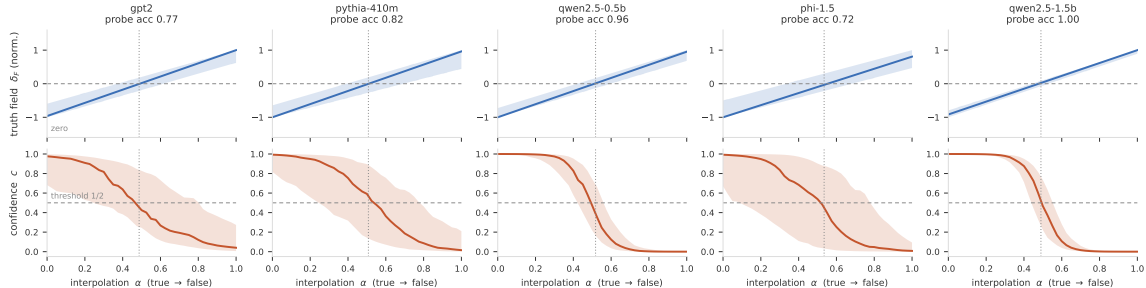


Figure 1: The predicted picture, observed. Top row shows the truth probe field along interpolation paths from a true statement to its negation, median and interquartile band over paths, normalized per path. Bottom row shows the independently trained confidence field on the same paths. The dotted vertical line marks the mean probe zero crossing. Confidence passes through its threshold of one half at the probe boundary in every model, most sharply in the strongest model.

Table 3: The model’s own confidence and null calibration. Decisive counts paths where the endogenous field separates the endpoints. Endo is the median crossing gap on decisive paths. Perm and Rand are median null gaps for permuted-label and random-direction heads over twenty seeds. Heads repeats the trained-head gap from Table 2.

Model	Decisive	Endo	Perm	Rand	Heads
GPT-2 (124M)	23/53	0.23	0.48	0.33	0.25
Pythia-410M	0/84	none	0.45	0.45	0.20
Qwen2.5-0.5B	128/137	0.13	0.43	0.40	0.10
Phi-1.5 (1.3B)	53/80	0.40	0.48	0.38	0.31
Qwen2.5-1.5B	146/147	0.08	0.45	0.45	0.05

Proceedings Track

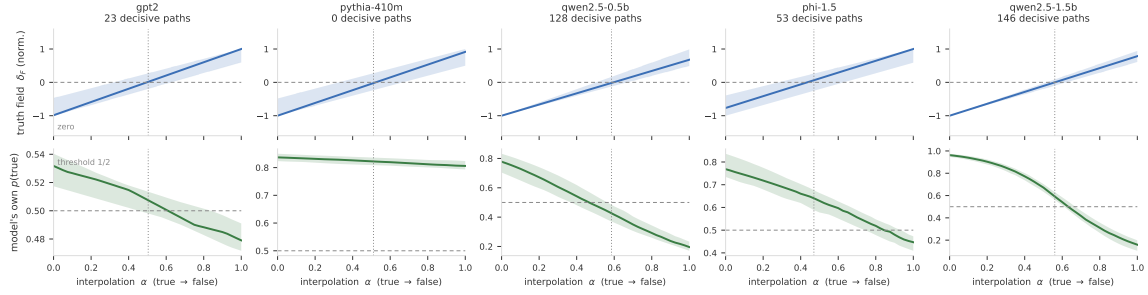


Figure 2: The model’s own confidence couples to the probe boundary where the separation premise holds. Rows as in Figure 1, but the bottom row is the model’s own probability of answering true, obtained by patching the interpolated final-block state, restricted to decisive paths. Qwen2.5-0.5B sweeps through one half at the probe boundary. Pythia-410M answers true everywhere, the premise fails, and no prediction follows. GPT-2 and Phi-1.5 sit between.

the final-verification file and by the files numbered 13 and 14. The experiment part includes the scripts, the public datasets of [Marks and Tegmark \(2024\)](#), the numerical results, and the figures. The theorems themselves depend on no experiment.

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